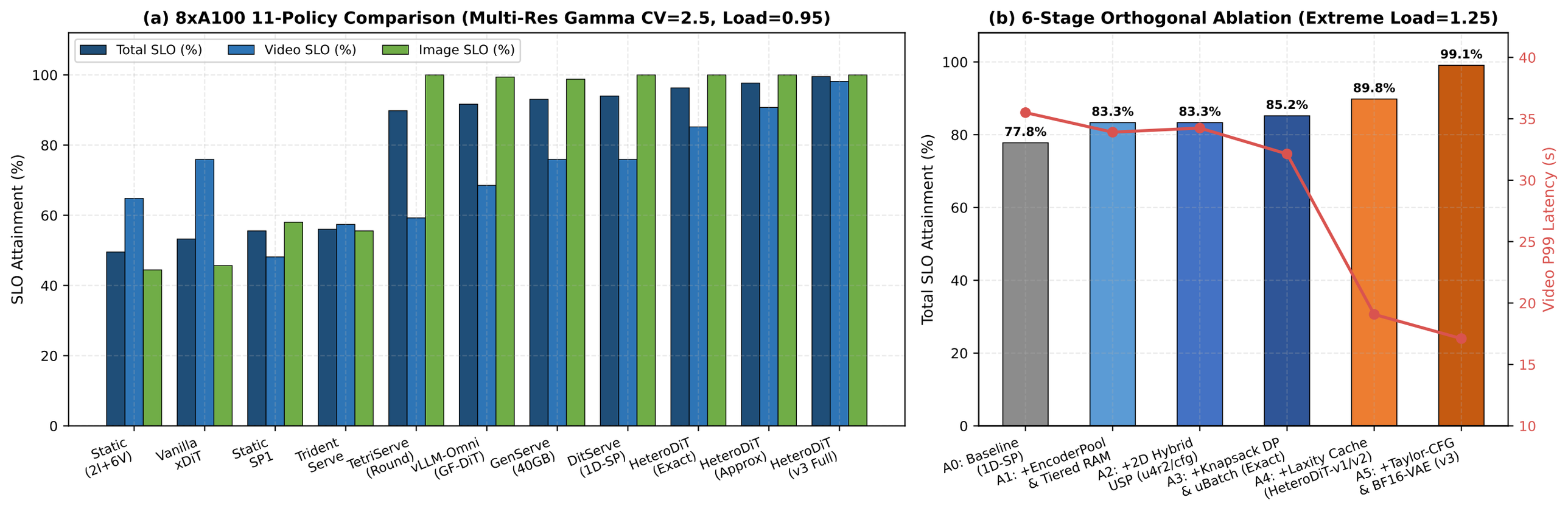
(a) 8xA100 11-Policy Comparison (Multi-Res Gamma CV=2.5, Load=0.95)
Total SLO (%)
Video SLO (%)
Image SLO (%)
SLO Attainment (%)
0
20
40
60
80
100
Static (2I+6V)
Vanilla xDiT
Static SP1
Trident Serve
TetriServe (Round)
vLLM-Omni (GF-DiT)
GenServe (40GB)
DitServe (1D-SP)
HeteroDiT (Exact)
HeteroDiT (Approx)
HeteroDiT (v3 Full)
(b) 6-Stage Orthogonal Ablation (Extreme Load=1.25)
Total SLO Attainment (%)
Video P99 Latency (s)
77.8%
83.3%
83.3%
85.2%
89.8%
99.1%
A0: Baseline (1D-SP)
A1: +EncoderPool & Tiered RAM
A2: +2D Hybrid USP (u4r2/cfg)
A3: +Knapsack DP & uBatch (Exact)
A4: +Laxity Cache (HeteroDiT-v1/v2)
A5: +Taylor-CFG & BF16-VAE (v3)